• In December 2016, total users = 3696 million, percentage of world population = 49.5

• In December 2017, total users = 4156 million, percentage of world population = 54.4

In IP next generation (IPng) or IP version 6 (IPv6), 128-bit address space (in place of 32 bits of IPv4) and minimum 40-byte IP header (in place of minimum of 20 bytes of IPv4) have been proposed. IPv6 is for coping with increasing demands of Internet access. IP addressing for IPv6 is with 128 bits or 16 bytes.

Second, due to the huge growth of Internet users, organisations are being assigned C addresses in IPv4. This is causing an exploration in the routing table of the Internet.

Third, IPv4 is for best effort delivery of packet. It neither guarantees packet sequence integrity nor bonded latency in delivery and is, hence, not suitable for real-time services, like voice and video. Thus, with IPv4, quality voice, video and multimedia services are not being serviced.

On the other hand, growth in voice IP traffic is increasingly huge. IPv6 has been proposed to address the stated limitations of IPv4. IPv6 does so with added features, as given below.

• 128-bit address size ensures address space of about 2^{128} compared to theoretical maximum of only 2^{32} in IPv4. IPv6 is believed to support trillions of networks.

• Provision for a single address associated with multiple interfaces.

• Auto-configuration facility.

• Provision for quality of service (QoS) support to real-time services like voice and video, and support to mobility.

• Flow level and priority in the header of IPv6 facilitates support of real-time data.

• Efficient header format (Fig. 1) having lesser overhead bits compared to IPv4.

• Fixed header format; header of IPv6 is different from that of IPv4 in many aspects, as follows:

(a) IPv6 has a fixed header size unlike IPv4. Options and padding are variable fields in IPv4. These have been removed in IPv6. This makes IPv6 act in asynchronous transfer mode (ATM) cell in comparison to IPv4, which is a conventional packet in data network.

Version (4 bits)	Priority (4 bits)	Flow label (24 bits)
Payload length (16 bits)	Next header (8 bits)	Hop limit (8 bits)
Source address (128 bits)		
Destination address (128 bits)		
Variable length TCP pack/UDP pack (which is TCP header + Payload or UDP header + Payload)...........		

Fig. 1: IPv6 packet format

(b) There is no header checksum in IPv6. This modification is made based on the experience that there is no need for checksum at each intermediate node.

(c) There is no hop-by-hop segmentation procedure in IPv6. Therefore there is no fragment offset, flags and identification fields in IPv6. There is also no need for header length field.

(d) Type of service (TOS) field of IPv4 is removed in IPv6. This removal is also based on experience, as TOS field is hardly used in IPv4.

• IPv6 defines six extension headers: hop-by-hop options, routing, fragment, encrypted security payload, authentication and destination options.

• IPv6 has different options, security and easy transition facility.

Technically, IPv6 can be compared with IPv4 in terms of two important parameters, namely, address space and IP header simplification. A few people projected the death of the Internet due to increased traffic, yet with IPv6, Internet access is guaranteed, with exponential increase in Internet users over time with the provision of abundant address space of 2^{128}.

With a simplified header description, IPv6 is prone to provide multimedia (voice, video and data) services. It is therefore the future replacement of IPv4.

It was felt that a new version after IPv4 should take care of more addresses, reduced overhead, better routing, support for address renumbering, improved header processing, reasonable security, and support for multimedia and mobility. IPv6 has two new fields: flow label and priority. These are included to facilitate the handling of real-time services like voice, video and quality multimedia services.

About IPv6

Salient features of IPv6, also shown in Fig. 1, may be summarised as:

• IPv6 has fixed header bytes like that in an ATM cell, which is of fixed 53 bytes. What is the advantage of having a fixed size? Let us consider the postal system. If envelopes are of variable sizes and weights, processing gets delayed, as the process includes measuring weight, verifying postal stamps required according to weight and so on. If there were fixed-sized and weighted envelopes, postal processing would take much less time and effort. Similarly, with a fixed header, node processing delay is reduced. This makes IPv6 more suitable for real-time services like voice, video and multimedia.

• A 128-bit address space in IPv6, instead of a 32-bit address space in IPv4, is believed to ensure that there will be no exhaustion of Internet address space. It is expected that the Internet under IPv6 supports 10^{15} (quadrillion) hosts and 10^{12} (trillion) networks. The Internet under IPv4 can support maximum 2^{32} hosts. IPv6 address space is about 2^{96} times more than that of IPv4. This is how it can ensure that exhaustion of address space is almost unlikely with IPv6.

• Besides unicast and multicast, IPv6 has the provision for anycast addressing that allows a packet addressed to an anycast address to be delivered to any one of the group of hosts.

• IPv6 has the provision for a single address being associated with multiple interfaces.

• Address auto configuration and